Energy Decay in a Damped Harmonic Oscillator

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October 2026

Abstract

We study the free motion of a mass on a linear spring subject to viscous damping. We derive the closed-form solution in the underdamped regime, show that the mechanical energy decays monotonically, and compare the analytic solution with two simple numerical schemes.

1 Introduction

The damped harmonic oscillator is the simplest model of a system that loses energy to its surroundings [1]. Its equation of motion is

$$m\ddot{x} + c\dot{x} + kx = 0, \tag{1}$$

where $m > 0$ is the mass, $c \geq 0$ the damping coefficient and $k > 0$ the spring constant. Dividing by m and writing $\omega_0 = \sqrt{\frac{k}{m}}$ and $\gamma = \frac{c}{(2m)}$ gives the *standard form*

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0. \tag{2}$$

2 Solution

Substituting $x = e^{\lambda t}$ into (2) gives the characteristic equation $\lambda^2 + 2\gamma\lambda + \omega_0^2 = 0$, whose roots are

$$\lambda_{\pm} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}. \tag{3}$$

The behaviour depends on the sign of the discriminant:

$$\text{regime} = \begin{cases} \text{underdamped}, \gamma < \omega_0, \\ \text{critically damped}, \gamma = \omega_0, \end{cases}$$

In the underdamped case let $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$. With $x(0) = x_0$ and $\dot{x}(0) = 0$ the solution is

$$x(t) = x_0 e^{-\gamma t}\left(\cos\omega_d t + \frac{\gamma}{\omega_d}\sin\omega_d t\right). \tag{4}$$

Equivalently, with $y = (x, \dot{x})^T$, equation (2)becomes the first-order system

$$\dot{y} = \begin{pmatrix} 0 & 1 \\ -\omega_0^2 & -2\gamma \end{pmatrix} y. \tag{5}$$

Theorem 1. *The energy $E(t) = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$ never increases. More precisely, $\frac{dE}{dt} = -c\dot{x}^2 \leq 0$.*

Proof. Differentiate E and use (1):: □

$$\frac{dE}{dt} = m\dot{x}\ddot{x} + kx\dot{x} \tag{6}$$

$$= \dot{x}(m\ddot{x} + kx) = -c\dot{x}^2. \tag{7}$$

3 Numerical check

We integrated (2)with $\omega_0 = 2, \gamma = 0.2$ and the classical fourth-order Runge–Kutta method (RK4) [2]. Nørsett. . For each step size h we:

- Euler's error is large because the method **adds** energy at every step, which works against the physical damping.

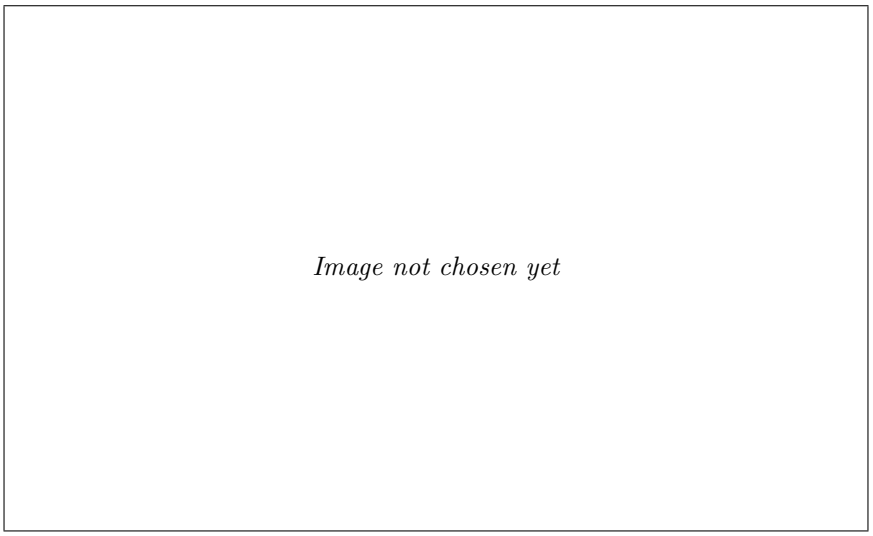


Figure 1: The exact solution for gamma = 0.2 and omega_0 = 2 and its envelope.Caption text.

1. integrated from t = 0 to t = 10 with 10/h steps;
2. compared every step with the exact solution (4);
3. recorded the largest absolute difference.

Table 1: Main results.

Quantity	**Value**	**Uncertainty**
a	1.00	0.02
b	2.50	0.05

4 Discussion

What the results mean, their limits, and what should come next.

5 Conclusion

The answer to the question in the introduction, in two or three sentences.

References

[1] H. Goldstein, C. Poole and J. Safko, *Classical Mechanics*, 3rd ed., Addison-Wesley, 2002.

[2] E. Hairer, S. P. Nørsett and G. Wanner, *Solving Ordinary Differential Equations I*, 2nd ed., Springer, 1993.